# TimeWeightedAgent

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## 1 Introduction

TimeWeightedAgent is a synchronous negotiation agent that improves upon AlmostEqualAgent. In SCML OneShot 2026, the shortfall penalty is set to be on average 5 to 10 times heavier than the disposal cost [1]. In this environment, TimeWeightedAgent focuses on securing quantity rather than improving prices to mitigate the risk of shortfall penalties. To achieve this, the offer strategy employs a "Safety Margin Strategy" and a "Stepwise Probabilistic Pricing" (described later) to secure profits while reducing the risk of shortfalls. Meanwhile, the acceptance strategy uses a "Fixed $\alpha$" to strongly prioritize quantity matching, acting as a bulwark to minimize our own penalties without being swayed by the opponent's proposals.

## 2 TimeWeightedAgent

### 2.1 Offer Strategy

As its offer strategy, TimeWeightedAgent calculates weights for each partner based on past transaction history to allocate proposed quantities, and uses stepwise probabilistic pricing based on time progression. The conventional allocation strategy adopted by AlmostEqualAgent had a static switching timing, making it difficult to adapt to recent changes in the opponent's behavior. Therefore, TimeWeightedAgent addresses this weakness and determines the proposed quantity and price through the following steps.

1. **Calculation of Reliability Score Based on Past Transaction History**

   We calculate the reliability score $W_i$ for each partner $i$ using an exponential time weighting ($\lambda = 2.0$) that highly evaluates transactions established later in the process:

$$W_i = \sum \left(Q_{\text{agreed}} \times \exp(\lambda t)\right) \tag{1}$$

Here, $Q_{\text{agreed}}$ is the quantity agreed upon in past transactions, and $t$ is the relative time when the negotiation was established ($0 \leq t \leq 1$).

2. **Target Quantity Considering Safety Margin**

   On the buyer's side, if the required quantity is strictly set as the target, a shortfall may occur due to some negotiation failures. Therefore, TimeWeightedAgent targets a quantity multiplied by a safety factor exclusively for the buyer's side.

$$Q_{\text{adjusted}} = \lfloor Q_{\text{need}}(1 + 0.20) \rfloor \tag{2}$$

   Here, $Q_{\text{need}}$ is the actual quantity the agent truly needs at that point in time.

3. **Allocation Based on Reliability Score**

   We prioritize partners with higher reliability scores $W_i$ as described above, and sequentially allocate quantities until the required quantity is reached within the bounds of the negotiation quantity constraints. This generates demands that consistently minimize our own penalties without being dragged down by the opponent's irregular proposed quantities.

4. **Stepwise Probabilistic Pricing**

   To induce the opponent's compromise while reliably securing the quantity, we perform probabilistic price offering according to the negotiation time. During counter-offers, we use the average time of all ongoing negotiations ($t_{\text{price}}$) as a baseline to avoid being swayed by a specific opponent's pace. Using the midpoint ($t = 0.50$) as a threshold, the probability $Pr(P_{\text{ideal}})$ of offering the ideal price $P_{\text{ideal}}$ is changed stepwise as follows:

$$Pr(P_{\text{ideal}}) = \begin{cases} 0.50 & (t_{\text{price}} < 0.50) \\ 0.25 & (t_{\text{price}} \geq 0.50) \end{cases} \tag{3}$$

   If $P_{\text{ideal}}$ is not selected, the most favorable concession price $P_{\text{concede}}$ for the opponent is offered. This realizes a flexible behavior that increases the probability of aiming for profit in the early stages, while suppressing aggressive demands to prioritize a reliable agreement (quantity securing) in the latter half.

## 2.2 Acceptance Strategy

In the acceptance strategy of TimeWeightedAgent, proposals from opponents are treated as a power set (all combinations of proposals). To suppress the risk of shortfall penalties to the utmost limit, we introduce a fixed $\alpha$ strategy that strongly prioritizes quantity matching.

1. **Scoring and Normalization**

We calculate the price score $P$ and the quantity matching score $Q$, normalizing each to a range of 0.0 to 1.0. Specifically, $P_{\text{score}} = \pm(P_{\text{total}} \mp P_{\text{penalty}})$ is normalized to $P_{\text{norm}}$, and $Q_{\text{score}} = |Q_{\text{all}} - Q_{\text{need}}|$ is normalized to $Q_{\text{norm}}$. Here, $P_{\text{total}}$ is the total price from the proposal combination, $P_{\text{penalty}}$ is the estimated shortfall/disposal penalty amount, and $Q_{\text{all}}$ is the total proposed quantity. For a buyer, the sum of the price and the penalty is treated as a negative evaluation, while for a seller, the revenue minus the penalty is treated as a positive evaluation, strictly avoiding our own penalties.

2. **Fixed $\alpha$ Evaluation**

   We set a weight $\alpha$ to emphasize quantity matching in the evaluation formula for acceptance candidates. In this study, we fix it at 0.85 based on preliminary experiments:

$$R_{\text{score}} = \alpha \times Q_{\text{norm}} + (1 - \alpha) \times P_{\text{norm}} \tag{4}$$

3. **Dynamic $\beta$ Threshold**

   For the acceptance decision, we use the most advanced negotiation time ($t_{\text{decision}}$) and lower the hurdle according to the lack of time remaining:

$$\beta_t = \beta_0(1 - 0.5t_{\text{decision}}) \tag{5}$$

   We select the combination with the highest $R_{\text{score}}$ from the power set, and accept it if it exceeds $\beta_t$.

# 3 Evaluation

In the experiments, we compare the scores of TimeWeightedAgent using AlmostEqualAgent, Rchan, and CostAverseAgent. These were selected as comparison targets because they are top-ranking agents in SCML OneShot 2025. The number of world environment configurations (n_configs) was set to 30. The experiments were conducted by setting the number of simulation days (n_steps) to 3 patterns: 50, 100, and 150. The score is used as the evaluation metric.

The scores are shown in Table 1. TimeWeightedAgent showed the highest average score among the comparison targets across all simulation day settings. These results suggest that strategically separating penalty avoidance and profit maximization is an effective approach.

Table 1: Experimental Results - Scores

| Agent | n_steps=50 | n_steps=100 | n_steps=150 |
|---|---|---|---|
| **TimeWeightedAgent** | **1.0535** | **1.0724** | **1.0661** |
| AlmostEqualAgent | 1.0471 | 1.0671 | 1.0630 |
| Rchan | 1.0407 | 1.0658 | 1.0613 |
| CostAverseAgent | 1.0364 | 1.0591 | 1.0445 |

## 4 Conclusion

TimeWeightedAgent performs time-weighted allocation and stepwise probabilistic pricing in offers, and evaluation using power sets in acceptance. We observed the shortcomings of past agents and added improvements to separate risk mitigation and profit maximization. As a result, it showed the highest average score among the comparison targets. Further verification is needed in the future regarding its effectiveness.

## References

[1] Y. Mohammed, A. Greenwald, K. Fujita, M. Klein, S. Morinaga, and S. Nakadai, *Supply Chain Management League (OneShot): Automated Negotiating Agents Competition*, SCML Organizing Committee, March 18, 2026.